

ZETALEV AB-55

ACID BUFFER — LEVELLING & MIGRATION AGENT FOR REACTIVE DYEING

Product Description:

Zetalev AB-55 is an acid buffer that plays a crucial role in ensuring reproducibility during the dyeing of reactive dyes on 100% cellulose fibres. Its buffering effect maintains consistent pH levels, making it particularly beneficial when dyeing polyester/cellulose blends with disperse and reactive dyes. It is suitable for use at all stages of processing in HT one-bath processes on jets and circulating liquor machines. Specifically, it stabilises the dyebath pH as temperatures rise, counteracting the tendency for pH to shift towards alkalinity during the migration process. It effectively arrests bicarbonates and carbonates in the bath liquor, preventing undesired pH shifts before the addition of strong alkali for dye fixation. This controlled pH environment facilitates even dye fixation and level dyeing, enhancing reproducibility. Additionally, Zetalev AB-55 possesses moderate dispersing and complexing properties, capable of sequestering positively charged earth metal ions present in the bath liquor, further contributing to process stability and quality outcomes.

Key Benefits:

- 1. Reproducibility Assurance:** Ensures consistent and reproducible dyeing processes, especially with reactive dyes on 100% cellulose fibres.
- 2. pH Stability Maintenance:** Maintains stable pH levels, crucial for dyeing polyester/cellulose blends with disperse and reactive dyes.
- 3. Bicarbonate & Carbonate Arrest:** Effectively arrests bicarbonates and carbonates in the bath liquor, preventing undesired pH shifts before alkali addition.
- 4. Dispersing & Complexing Properties:** Moderate dispersing and complexing action sequesters positively charged earth metal ions for improved process stability.

Properties:

Chemical Constitution	Formulated with carboxylic acid derivatives
Ionic Character	Anionic
pH of 1% solution	Approx. 5
Specific Gravity at 20°C	Approx. 1 gm/cc
Physical Form	Liquid
Storage Stability	Stable for 1 year at 20°C; not affected by temperature fluctuations
Stability to Alkalis	Not applicable
Stability to Acids	Very Good
Stability to Oxidants	Very Good
Stability to Hard Water	Very Good
Compatibility	Compatible with both non-ionic and anionic products

Applications & Dosage:

Zetalev AB-55 must be added to the dye bath at the start of the HT exhaust dyeing cycle, before the dyes. It can be diluted with water in all proportions.

Recommended Dosage — PES/CEL blends and CEL fibres:

0.5 to 2.0 GPL in the dyebath. Exact dosage depends on dye quantity, water composition, fabric characteristics, salt conditions, liquor ratio, and other chemicals present. Determine precise dosage through preliminary trials for best results.

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